

① Estimate the Effort (person-months) and development time (months)

A software project is estimated to have a size of 40 KLOC and is developed in organic mode using the basic cocomo model.

Given mode = organic

$$\text{size} = 40 \text{ KLOC}$$

$$a = 2.4$$

$$b = 1.05$$

$$c = 2.5$$

$$d = 0.38$$

$$\text{Effort, } E = a \times (\text{KLOC})^b = 2.4 \times (40)^{1.05}$$

$$\therefore E = 2.4 \times 46.2$$

$$E = 110.88 \text{ person-months}$$

$$\text{development time, } T = c \times (E)^d$$

$$T = 2.5 \times (110.88)^{0.38}$$

$$T = 2.5 \times 5.97$$

$$T = 14.92 \text{ months}$$

② A software project of size 55 KLOC is developed in semi-detached mode using the intermediate cocomo model, the Effort Adjustment Factor (EAF) = 1.10. calculate the Effort and development time.

Given, mode = semi-detached.

$$\text{size} = 55 \text{ KLOC}$$

$$\text{EAF} = 1.10$$

$$a = 3.0, b = 1.12$$

$$c = 2.25$$

$$d = 0.35$$

$$\text{Effort, } E = a \times (1000)^b \times \text{EAF}$$

$$= 3 \times (55)^{1.12} \times 1.10$$

$$= 3 \times 72.3 \times 1.10$$

$$E = 238.59 \text{ person-months}$$

$$\text{development time, } T = c \times (E)^d$$

$$= 2.5 \times (238.59)^{0.35}$$

$$= 2.5 \times 6.81$$

$$T = 17.02 \text{ months}$$